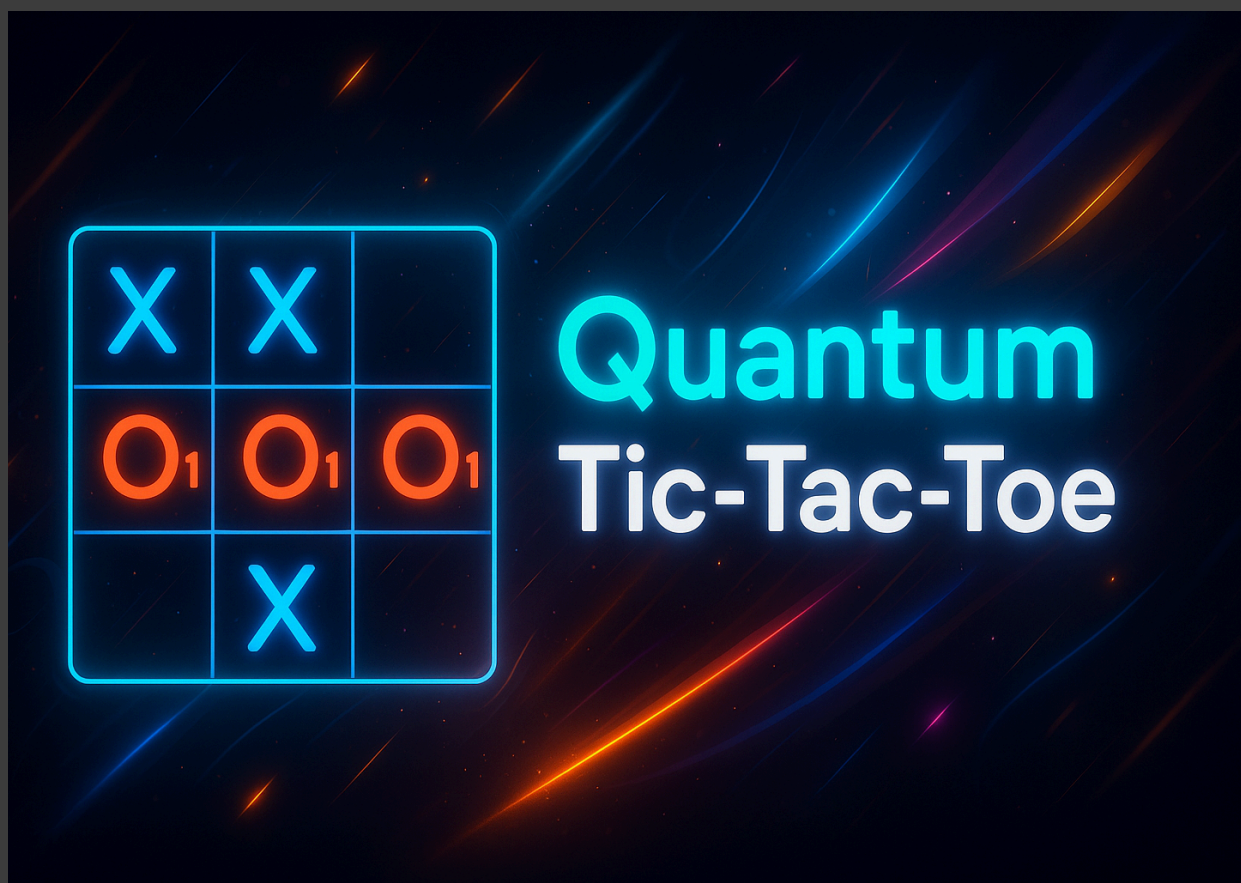


Quantum Tic-Tac-Toe

Report on the Project for CQ Hacks Hackathon



Group Members

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CQ Hacks

PROJECT OVERVIEW

The **Quantum Tic-Tac-Toe** project is an advanced web-based game that combines the classic game of Tic-Tac-Toe with core principles of quantum computing. It offers an interactive and educational platform that introduces users to quantum mechanics through gameplay, featuring quantum gate animations, Simple AI opponent, verification quizzes, and ultra-futuristic UI.

The game is designed for students and beginners with no prior background in quantum mechanics. It allows users to explore ideas such as superposition, collapse, entanglement, and quantum gates through a simple interface and hands-on interaction.

OBJECTIVES

- Gamify quantum computing concepts.
- Provide an engaging way to understand quantum gates (Hadamard, CNOT, etc.).
- Test user understanding with MCQs and voice input.
- Showcase modern web development through visual effects and animation.

TECHNOLOGY USED

1. **Frontend:** HTML, CSS, JavaScript.
2. **UI Enhancements:** SVG animations, CSS keyframes, JS modal controls.
3. **Voice Input:** Web Speech API.
4. **Local Storage:** Used for saving theme preferences and game state.

KEY FEATURES

Game Mechanics:

- Classic Tic-Tac-Toe rules with a twist: quantum behaviors and AI moves.
- Hover preview and animated tile transitions.
- AI Mode for single-player experience using basic strategy.

Quantum Integration:

- Quantum Gate Animations: SVG/CSS-based dynamic visuals for Hadamard, CNOT, etc.
- Animated Quantum Gate Intro Modal: Eye-catching entry point to introduce the concept.
- Verification system includes:
 - ❖ Randomized MCQ Quiz about quantum principles.
 - ❖ Voice Input Support for user answers.
 - ❖ "Learn More" Button for deeper understanding via linked articles.

Design and UI:

- Ultra-futuristic theme with:
 - Particle background
 - Neon glow effects
 - Dark mode toggle
 - Animated transitions
- Responsive layout for all screen sizes.

ACHIEVEMENTS

- Created a playable quantum-inspired game.
- Delivered immersive UI/UX aligned with sci-fi and quantum aesthetics.
- Integrated basic educational content into entertainment.

SCOPES FOR FUTURE ENHANCEMENTS

- Add **real quantum randomness** via IBM Qiskit API or Quantum Random Number Generators.
- Implement multiplayer mode.
- Store user performance and quiz results.
- Add leaderboard and level-based quizzes.
- Introduce quantum superposition in gameplay (e.g., double marks until measured).
- Access the big model API to generate more diverse pre-questions.

CONTRIBUTIONS

SHREYAS KHARAT: Frontend (HTML, CSS, JAVASCRIPT)

ZESHAN ZHANG: UI Enhancements (SVG animations, CSS keyframes, JS modal controls)

DEVELOPER'S NOTE

This project reflects the intersection of science and code. It's designed not only to entertain but also to ignite curiosity about quantum computing.

CONCLUSION

The **Quantum Tic-Tac-Toe** project stands as a unique fusion of entertainment, education, and futuristic design. By embedding quantum logic into a familiar game, it transforms a simple pastime into an interactive learning tool. Through features like AI gameplay, animated quantum gate visuals, and voice-interactive quizzes, users are not just playing — they are exploring the fundamental ideas of quantum computing in an engaging, gamified way. This project exemplifies how complex scientific concepts can be simplified and made accessible through creative development and immersive design. It's not just a game — it's a gateway into the quantum world.

LINK TO PROJECT

Get our project on GitHub Repo:

<https://github.com/shreyaskharat19/Quantum-Tic-Tac-Toe>

LINK TO OUR GITHUB ACCOUNTS

Shreyas Kharat:

GitHub Account: <https://github.com/shreyaskharat19>

Zeshan Zhang:

GitHub Account: <https://github.com/zhangmask>

IMAGE OF THE INTERFACE

Main Game Interface:

